

The sharp constant for rank-one coordinate maps and unconditional frame rescaling

Codex agent lane 11

July 21, 2026

Status. Full result likely valid, subject to human review. The optimal complex constant in Tselishchev's Theorems 2.2 and 2.3 is $4/\pi$; the optimal real rescaling constant is $\pi/2$.

Source problem

Tselishchev proves that there is a universal K such that every bounded map $\Phi : \ell_\infty^n \rightarrow B(H)$ with rank-one coordinate images satisfies $\|\Phi\|_{\text{cb}} \leq K\|\Phi\|$, and proves the equivalent finite rescaling theorem for nonzero vector pairs [3, Theorems 2.2–2.3]. His proof gives $K = 2$. Remark 2.6(1), on PDF page 5, says that Gaussian variables improve the constant and identifies evaluation of the best possible K as an interesting problem.

(strictly speaking, for nonzero vectors $\{x_k\}_{k=1}^\infty$ and $\{y_k\}_{k=1}^\infty$) holds with the same constant.

2.6. Remarks. 1. The evaluation of the best possible constant K in Theorems 2.2 and 2.3 is beyond the scope of the present paper, although it seems to be an interesting problem. We only mention that by using the analogue of Khintchine's inequality for independent Gaussian random variables (as it is done in [Pis12, Section 5]) one can get the better constant K than $K = 2$ which is obtained in our proof below.

The frame-multiplier formulation originated in Balazs–Freeman–Popescu–Speckbacher [1, Conjecture 1.2]. We solve the sharp-constant problem in both of Tselishchev's finite formulations.

Definitions and main theorem

For complex Hilbert spaces, let K_{cb} be the least K such that

$$\|\Phi\|_{\text{cb}} \leq K\|\Phi\|$$

for all finite-dimensional H , all n , and all $\Phi : \ell_\infty^n \rightarrow B(H)$ satisfying $\text{rank } \Phi(e_k) \leq 1$.

Let $K_{\text{res}}^{\mathbb{C}}$ be the least K in the following nonzero-vector rescaling statement. Whenever

$$\left\| \sum_{k=1}^n \varepsilon_k m_k \langle u, y_k \rangle x_k \right\| \leq C\|u\| \quad (u \in H, |\varepsilon_k| = 1), \quad (1)$$

with $m_k \neq 0$ and $x_k, y_k \neq 0$, there are factors $c_k \overline{d_k} = m_k$ for which both $(c_k x_k)$ and $(d_k y_k)$ have Bessel bound at most KC . Define $K_{\text{res}}^{\mathbb{R}}$ analogously over real Hilbert spaces, using signs.

Theorem 1 (Sharp constants). *The optimal constants are*

$$K_{\text{cb}} = K_{\text{res}}^{\mathbb{C}} = \frac{4}{\pi}, \quad K_{\text{res}}^{\mathbb{R}} = \frac{\pi}{2}.$$

In particular, the Gaussian improvement suggested in Remark 2.6(1) is sharp in both the completely bounded-map and rescaling formulations.

Proof intuition

The upper bound comes from the exact first absolute Gaussian moment. The operator norm of a rank-one coordinate map yields an ℓ_1 estimate for all scalar matrix coefficients. Applying that estimate to two independent circular Gaussian sums turns the d -level matrix coefficient into a sum of rank-one trace norms and costs exactly $(\mathbb{E}|\gamma|)^{-2} = 4/\pi$.

For rescaling, the same Gaussian calculation bounds a Hellinger-type sum for every pair of density matrices. A convex minimization over logarithmic weights balances the two frame operators coordinate by coordinate, making both Bessel bounds equal to that controlled sum.

Sharpness comes from independent Haar unit vectors. In dimension d , the ordinary norm of the associated empirical rank-one map approaches the square of the first absolute coordinate moment. Regardless of the sample, a d -level amplification is at least $1/d$. The quotient tends to $4/\pi$. The same samples obstruct better Bessel rescaling by taking traces of the two frame operators.

The sharp upper bounds

Lemma 1 (Complex completely bounded upper bound). *If $\Phi : \ell_\infty^n \rightarrow B(H)$ has rank at most one on every coordinate, then*

$$\|\Phi\|_{\text{cb}} \leq \frac{4}{\pi} \|\Phi\|.$$

Proof. Write

$$\Phi(a)u = \sum_{k=1}^n a(k) \langle u, y_k \rangle x_k, \quad C = \|\Phi\|.$$

Taking scalar matrix coefficients and aligning phases gives

$$\sum_{k=1}^n |\langle u, y_k \rangle| |\langle x_k, v \rangle| \leq C \|u\| \|v\|. \quad (2)$$

Fix m and vectors $u_1, \dots, u_m, v_1, \dots, v_m \in H$. Substitute $u = \sum_j \gamma_j u_j$ and $v = \sum_i \delta_i v_i$ in (2), where the two families are independent standard circular complex Gaussians, normalized by $\mathbb{E}|\gamma|^2 = 1$. Since

$$\mathbb{E}|\gamma| = \frac{\sqrt{\pi}}{2},$$

independence and Cauchy–Schwarz give

$$\sum_{k=1}^n \left(\sum_j |\langle u_j, y_k \rangle|^2 \right)^{1/2} \left(\sum_i |\langle x_k, v_i \rangle|^2 \right)^{1/2} \quad (3)$$

$$\leq \frac{4}{\pi} C \left(\sum_j \|u_j\|^2 \right)^{1/2} \left(\sum_i \|v_i\|^2 \right)^{1/2}. \quad (4)$$

An element of $M_m(\ell_\infty^n)$ is a tuple $(A_k)_{k=1}^n$ with norm $\max_k \|A_k\|$. In a matrix coefficient of the amplification, the term belonging to k is the trace pairing of A_k with the rank-one matrix

$$B_k = (\langle u_j, y_k \rangle \langle x_k, v_i \rangle)_{i,j}.$$

Its trace norm is precisely the k th product on the left side of (3). Trace duality proves

$$\|\Phi_{(m)}((A_k))\| \leq \frac{4}{\pi} C \max_k \|A_k\|.$$

Taking the supremum over m proves the lemma. $\square$

Lemma 2 (Gaussian rescaling upper bound). *The complex and real nonzero-vector rescaling constants satisfy*

$$K_{\text{res}}^{\mathbb{C}} \leq \frac{4}{\pi}, \quad K_{\text{res}}^{\mathbb{R}} \leq \frac{\pi}{2}.$$

Proof. Absorb the symbols and their phases into nonzero vectors X_k, Y_k so that (1) becomes

$$\left\| \sum_k \varepsilon_k \langle u, Y_k \rangle X_k \right\| \leq C \|u\|. \quad (5)$$

Put $P_k = X_k \otimes X_k$ and $Q_k = Y_k \otimes Y_k$. For density matrices $\rho, \sigma \in \mathcal{D}(H) := \{R \geq 0 : \text{Tr } R = 1\}$, apply the scalar consequence of (5) to two independent Gaussian vectors with covariances σ and ρ . Exactly as above,

$$\sum_k \sqrt{\text{Tr}(\rho P_k) \text{Tr}(\sigma Q_k)} \leq L, \quad (6)$$

where $L = 4C/\pi$ in the complex case. For real standard Gaussians, $\mathbb{E}|g| = \sqrt{2/\pi}$, so $L = \pi C/2$.

For $s \in \mathbb{R}^n$, define the convex functions

$$A(s) = \lambda_{\max} \left(\sum_k e^{s_k} P_k \right), \quad B(s) = \lambda_{\max} \left(\sum_k e^{-s_k} Q_k \right).$$

Because all P_k, Q_k are nonzero, $A + B$ is coercive and has a minimizer s^* . The variational formula for the top eigenvalue and the subgradient optimality condition give maximizing density matrices ρ, σ such that, for every k ,

$$e^{s_k^*} \text{Tr}(\rho P_k) = e^{-s_k^*} \text{Tr}(\sigma Q_k). \quad (7)$$

Indeed, the subgradients of A are convex combinations of the vectors $(e^{s_k} \text{Tr}(\rho P_k))_k$ from maximizing states; their convex combinations are again represented by a maximizing density matrix, and similarly for B .

Writing $a_k = \text{Tr}(\rho P_k)$ and $b_k = \text{Tr}(\sigma Q_k)$, equation (7) and (6) yield

$$A(s^*) = \sum_k e^{s_k^*} a_k = \sum_k \sqrt{a_k b_k} \leq L, \quad B(s^*) \leq L.$$

Thus $\alpha_k = e^{s_k^*/2}$ makes $(\alpha_k X_k)$ and $(\alpha_k^{-1} Y_k)$ both L -Bessel. Restoring the absorbed symbols proves the lemma. $\square$

Finite Haar examples

Let $S_{\mathbb{F}}^{d-1}$ be the unit sphere of $\mathbb{F}^d$, with Haar probability measure, for $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$. Define

$$h_{\mathbb{F},d} = \mathbb{E}|\langle e_1, Z \rangle|, \quad Z \sim \text{Haar}(S_{\mathbb{F}}^{d-1}).$$

Lemma 3 (Haar moments). *The exact moments are*

$$h_{\mathbb{C},d} = \frac{\Gamma(d)\Gamma(3/2)}{\Gamma(d+1/2)}, \quad dh_{\mathbb{C},d}^2 \longrightarrow \frac{\pi}{4}, \quad (8)$$

$$h_{\mathbb{R},d} = \frac{\Gamma(d/2)}{\sqrt{\pi}\Gamma((d+1)/2)}, \quad dh_{\mathbb{R},d}^2 \longrightarrow \frac{2}{\pi}. \quad (9)$$

Proof. In the complex case, $|\langle e_1, Z \rangle|^2$ has the Beta(1, $d-1$) distribution. Integrating its square root gives the first formula. In the real case, the corresponding distribution is Beta(1/2, $(d-1)/2$), giving the second formula. The limits follow from $\Gamma(t+a)/\Gamma(t+b) \sim t^{a-b}$. $\square$

Lemma 4 (Finite empirical realization). *For every fixed d , every $\eta > 0$, and either scalar field, there are finitely many pairs of unit vectors $(x_k, y_k)_{k=1}^N$ such that*

$$\sup_{\|u\|=\|v\|=1} \frac{1}{N} \sum_{k=1}^N |\langle u, y_k \rangle| |\langle x_k, v \rangle| \leq h_{\mathbb{F},d}^2 + \eta. \quad (10)$$

Proof. Take independent Haar pairs (X_k, Y_k) . For fixed unit u, v , the expected summand factors and equals $h_{\mathbb{F},d}^2$. The functions

$$(x, y) \longmapsto |\langle u, y \rangle| |\langle x, v \rangle|$$

are uniformly bounded and are Lipschitz in (u, v) with a common constant. Choose a finite δ -net of the compact product of unit spheres, apply the law of large numbers simultaneously on the net, and then let δ be small. Some finite sample satisfies (10). $\square$

For such a sample in the complex case, define

$$\Phi_N(a)u = \frac{1}{N} \sum_{k=1}^N a(k) \langle u, y_k \rangle x_k. \quad (11)$$

Aligning phases shows that the left side of (10) is exactly $\|\Phi_N\|$. Hence

$$\|\Phi_N\| \leq h_{\mathbb{C},d}^2 + \eta. \quad (12)$$

Lemma 5 (Amplification lower bound). *For every collection of complex unit-vector pairs in $\mathbb{C}^d$, the map (11) satisfies*

$$\|\Phi_N\|_{\text{cb}} \geq \|\Phi_{N,(d)}\| \geq \frac{1}{d}.$$

Proof. Let $(e_j)_{j=1}^d$ be the standard basis and put $u_j = v_j = e_j/\sqrt{d}$. These define unit vectors in the Hilbert direct sum $(\mathbb{C}^d)^d$. For each k , the rank-one coefficient matrix

$$B_k = (\langle u_j, y_k \rangle \langle x_k, v_i \rangle)_{i,j=1}^d$$

has trace norm

$$\|B_k\|_1 = \left(\sum_j |\langle u_j, y_k \rangle|^2 \right)^{1/2} \left(\sum_i |\langle x_k, v_i \rangle|^2 \right)^{1/2} = \frac{1}{d}.$$

Choose $A_k \in M_d$ with $\|A_k\| \leq 1$ whose trace pairing with B_k (with the harmless transpose dictated by conventions) equals $\|B_k\|_1$. Then $A = (A_k) \in M_d(\ell_\infty^N)$ has norm at most one, while the matrix coefficient of $\Phi_{N,(d)}(A)$ between (u_j) and (v_i) equals

$$\frac{1}{N} \sum_{k=1}^N \|B_k\|_1 = \frac{1}{d}.$$

□

Lemma 6 (Rescaling lower bound). *Regard the sample as a multiplier with symbols $m_k = 1/N$. If $c_k \overline{d_k} = 1/N$ and the sequences $(c_k x_k)$ and $(d_k y_k)$ have Bessel bounds B_x, B_y , then*

$$\max\{B_x, B_y\} \geq \frac{1}{d}.$$

The same statement holds over the real field.

Proof. Taking traces of the two frame operators gives

$$\sum_k |c_k|^2 \leq dB_x, \quad \sum_k |d_k|^2 \leq dB_y,$$

because all x_k, y_k are unit vectors. Therefore

$$1 = \sum_k |c_k d_k| \leq \left(\sum_k |c_k|^2 \right)^{1/2} \left(\sum_k |d_k|^2 \right)^{1/2} \leq d \sqrt{B_x B_y}.$$

□

Sharpness and conclusion

For the complex samples, Lemmas 4 and 5 give

$$\frac{\|\Phi_N\|_{\text{cb}}}{\|\Phi_N\|} \geq \frac{1}{d(h_{\mathbb{C},d}^2 + \eta)}.$$

First let $\eta \downarrow 0$ and then $d \rightarrow \infty$. By (8), this proves $K_{\text{cb}} \geq 4/\pi$. Together with Lemma 1,

$$K_{\text{cb}} = \frac{4}{\pi}.$$

For any rescaling of the same multiplier, Lemma 6 forces a Bessel bound of at least $1/d$, while its unconditional constant is at most $h_{\mathbb{C},d}^2 + \eta$. Thus $K_{\text{res}}^{\mathbb{C}} \geq 4/\pi$. Lemma 2 gives equality.

Repeating the empirical construction on the real sphere, using (9) and the real part of Lemma 6, gives $K_{\text{res}}^{\mathbb{R}} \geq \pi/2$. The real upper bound in Lemma 2 again gives equality. This completes the proof of Theorem 1.

Verification notes and scope

- Every sharpness example is finite-dimensional and has nonzero vectors and nonzero symbols, so it belongs to the exact nondegenerate formulation of Tselishchev’s Theorem 2.2.
- The lower bound requires no numerical simulation. Random sampling is used only as a probabilistic existence proof for a finite family; uniformity follows from compactness and a finite net.
- The amplification witness is at level d and is explicit once the sample is fixed. Each matrix A_k is a polar trace-duality witness for a rank-one coefficient matrix.
- If one retains irrelevant indices with a zero vector and a nonzero symbol, the exact $4/\pi$ bound need not be attained by finite factors; every constant strictly larger than $4/\pi$ still works. The source’s quantitative Theorem 2.2 explicitly uses the nonzero-vector formulation.
- This packet determines the rank-one constant only. The growth of the rank- r constants $K(r)$ from Remark 2.6(2) remains open.

Bounded novelty check

The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2508.02802, “rank-one completely bounded”, “best constant”, and “ $4/\pi$ ”. The local 2000–present arXiv source corpus was searched for the same combinations. Exact web searches on July 21, 2026 included the source title and author, both symbolic forms of $4/\pi$, rank-one coordinate maps, Haar unit-vector examples, and forward references to arXiv:2508.02802. Pisier’s account of the little Grothendieck theorem [2, Section 5] was checked as the Gaussian background explicitly cited by the source. No source stating the sharp constant for Tselishchev’s two formulations was found; the source itself says the best constant is beyond its scope.

The mechanism is closely related to classical sharp Gaussian constants, so the recommended human review should particularly check whether this precise rank-one-map specialization has appeared under different operator-ideal or tensor-norm terminology. Mathematically, the main review points are the uniform empirical realization and the trace-pairing amplification witness.

References

- [1] P. Balazs, D. Freeman, R. Popescu, and M. Speckbacher, *Quantitative bounds for unconditional pairs of frames*, arXiv:2212.00947 (2022).
- [2] G. Pisier, *Grothendieck’s theorem, past and present*, Bull. Amer. Math. Soc. (N.S.) **49** (2012), 237–323; arXiv:1101.4195.
- [3] A. Tselishchev, *Rescaling of unconditional Schauder frames in Hilbert spaces and completely bounded maps*, arXiv:2508.02802 (2025).